Unsupervised-learning financial reconciliation: a robust, accurate approach inspired by machine translation

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Financial reconciliation (cross-checking independent sources of data) is a time-honored and widespread function in finance and audit. Its objectives are to ensure completeness, timeliness, and accuracy of recording of transactions.

With the proliferation of data over recent decades, the ways in which reconciliation is approached have evolved. There are now a number of software products that promise to automate the process of detailed transaction matching. However, the 'state of the art' in these products is rules-based reconciliation. Rules-based systems in any domain tend to be brittle in that any change in the data streams causes the rules to have to be debugged and rebuilt, itself often a time-consuming process. And to make matters worse, this might be caused not just by data schema changes, but the contents of what is in the data fields themselves. Either of these can occur if a third party such as a bank changes its internal processes.

In a sense, automated reconciliation is where machine translation was almost 70 years ago; IBM's 1954 Georgetown experiment approached Russian-to-English translation using rules, but it took another 4 decades for researchers in data-driven Artificial Intelligence (AI) to realize why machine translation did not initially live up to its promises and develop a truly robust methodology for machine translation based on unsupervised learning.

It turns out that financial reconciliation can be cast as a machine translation problem based on unsupervised learning. To our knowledge, we are the first to propose this. Here, we demonstrate via experiments on real-life (albeit small-scale) financial data that this way of approaching the problem demonstrates promise in terms of accuracy, as well as solving the problem of lack of robustness inherent in rules-based approaches.

CCS CONCEPTS • Information systems • human-centered computing • applied computing

Additional Keywords and Phrases: computing methodologies, unsupervised learning, reconciliation, machine translation

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1 BACKGROUND

Financial reconciliation is a type of internal control used to increase assurance of completeness, timeliness and accuracy in financial information [1]. It has been a staple of internal and external audit since the ancient world, a type of (dishonest) reconciliation appearing even in one of Jesus's parables (Luke 16:1-8). There is a reason for this: the premise of reconciliation is to compare data about a single set of facts from independent sources – for example, the bank's view of one's bank account versus one's own records – and as the AICPA's auditing standard AU-C 500.A8 states, 'corroborating information obtained from a source independent of the entity may increase the assurance that the auditor obtains from audit evidence that is generated internally' [2].

In a sense, none of this has changed with the advent of the age of 'big data'. What has changed is scale and electronic availability of data sources. With larger-scale datasets, it is increasingly infeasible on finite and limited budgets to perform by hand the kind of comparisons detailed reconciliation demands. But this challenge has been offset by opportunities that electronic availability affords. Now, there are a variety of off-the-shelf software products, including enterprise-level products, that promise to automate reconciliation (see https://softwareconnect.com/bank-reconciliation/ for a recent [December 2019] example reviewing various offerings).

However, a new problem is emerging, one which is still under-appreciated. The state of the art in these software systems, implicitly acknowledged in the above review, is that such systems function based on 'matching rules specific to one account or company-wide' (see https://www.blackline.com/transaction-matching/ [retrieved 4/2/2020] for one example). What is wrong with that? What is overlooked is the maintenance that will be required over time. Rules require exceptions to deal with anomalous data that does not fit the overall pattern. Exceptions increase the complexity of the system – not necessarily just the software per se, but the larger system which includes 'humans in the loop' and in which the software plays a part. When that larger system produces unexpected results, as it inevitably will when data formats change (or even when data formats do not change, but the patterns of data change within existing data formats), an effort must be undertaken to debug the system, and in many cases, rewrite the rules. The burden of doing this, with rules-based software, is likely to fall on the end user. It is almost an axiom that automation saves time, but when it does go wrong and users have come to rely on it, it goes spectacularly wrong.

The world of finance is actually following a path here that was trodden many decades ago in another field, the field of machine translation. During the Cold War, the U.S. national security apparatus funded research into machine translation, and this resulted in IBM's 'Georgetown Experiment' in 1954 [3][4]. In this experiment, sixty Russian sentences were successfully translated into English, and the researchers involved asserted that machine translation would be a solved problem by the end of the 1950s, a claim that some in the scientific community treated with skepticism at the time [5]. Sure enough, by the early 1960s, U.S. government sponsors had become frustrated with the lack of progress: the six rules in the Georgetown Experiment simply did not scale up to larger and more diverse datasets, and this was not a problem that it was computationally feasible to solve, in the end, simply by adding more rules [5].

For several decades, machine translation languished until availability of 'big data' made Statistical Machine Translation (SMT) possible [6]. The premise of SMT, which (in contrast to the earlier rules-based systems) is based on unsupervised machine learning, is to learn from parallel corpora – that is, examples of already-translated material. Unlike IBM's system for which more data was its downfall, SMT performs better the more data is given to it, as it leverages machine learning to mimic what humans have already done. The more data,

the more likely it is – by the law of large numbers – that particular patterns requiring translation will have been encountered before, and the more likely that the machine can use subtle patterns of context to determine the translation appropriate in the given circumstances. Furthermore, unlike the Georgetown system (which was built specifically for Russian-to-English translation), the same probabilistic algorithms inherent in SMT can be applied with virtually no modification to any language pair. Naturally, it follows from this that 'noise' in linguistic data, such as speaker-specific variation or changes in usage, can be taken in stride as long as the 'noise' is within the overall spectrum of available existing data.

This brings us back full circle to financial reconciliation. The contribution of this paper is to frame reconciliation as a 'machine translation' problem, something we believe has not been done before in scientific literature, and show in practice how this can be implemented and that it requires no heuristics specific to a given dataset (section 2), and how accurate and robust it is, empirically (section 3). In section 4 we will conclude by saying that this demonstration, albeit on a dataset of relatively small scale, yields results that show promise and, in theory and based on the prior example of SMT, scalability to datasets of much larger size.

2 RECONCILIATION AS 'MACHINE TRANSLATION'

2.1 Machine learning mechanisms

Recall from section 1 that we stated that the essence of reconciliation is to compare two independent datasets bearing witness to the same set of financial events (transactions). A common example of a reconciliation is a bank reconciliation: comparison of a check register (one's own records) to a bank statement (the bank's records). One can think of this in linguistic terms: the bank describes the transactions in one 'language', while the check register describes the same transactions in another. Following this analogy, there is a single 'sentence' of the bank's 'language' for each transaction. The 'words' in the sentence might consist of a transaction date, the actual words included in the transaction description, a transaction type, a check number, and other information. Similarly, each entry in the check register is like a 'sentence', though the 'words' may differ from the bank's. The bank's language might even be 'cognate' to that of the check register in that words written identically have similar, but not quite identical, meanings for non-coincidental reasons (think English versus French 'grand'). It might also be that there is no overlap between the 'words' used by the bank and the check register – a scenario that could completely defeat rules-based reconciliation (think Russian versus English, which are written in different scripts). Now, taking the analogy of languages further and extending it into the realm of SMT, we can liken the bank statement and check register taken together to a 'parallel corpus'. The 'sentences' are not just random, they can actually be aligned in parallel, just as one can align a Russian and English sentence that have the same meaning. One way this alignment could happen is through previously seen examples (those where a human has matched individual bank transactions to check register transactions), just as SMT uses previously-translated text as the basis to learn its patterns. The key with SMT is that there are no 'rules'; translation simply improves on the basis of more examples.

Now we have taken the analogy this far, it should become apparent that reconciliation can be treated as essentially another SMT problem. Whether we are dealing with 'human' language or 'financial' language is beside the point, because both are simply data, and SMT is data-driven. The basic mechanisms already exist within SMT to 'translate' as needed. For example, if dates do not exactly match because they are consistently

recorded a few days earlier in the check register than they are by the bank, this can be handled probabilistically by the algorithms within SMT.

In fact, we have used SMT to demonstrate the concepts thus far, because SMT has at least some familiarity to a wider audience, but the problem of automated reconciliation is even more akin to the related field of cross-language information retrieval [7][8], the purpose of which is to retrieve the most relevant documents in language X given a user query in language Y. In the case of reconciliation, an automated matching algorithm may be set up to 'retrieve' the 'most relevant' or 'most similar' transaction from the check register for each transaction from the bank statement (or vice versa), and reconcile them. Retrieval is, in fact, a somewhat easier problem than translation, because it involves simple recognition (of existing patterns) rather than generation (for example, of new text). However, some of the basic machine learning principles and mechanisms are the same as in SMT: the algorithms can be set up to learn from a parallel multilingual corpus, and will tend to perform better the more data they are given.

In automated financial reconciliation, we have one advantage available from a machine learning point of view that is not usually available in machine translation, which is that sometimes it is possible to make a good educated guess as to which transactions 'go together' even without human intervention, and even to estimate the probability based on the data alone that such matches could have happened by chance (giving a statistically justifiable way to avoid spurious matches). This is because many amounts may be unique within the entire dataset, and if there happens to be a match on the same unique amount between the check register and bank statement, there may be very good grounds to postulate a match even in the absence of any other information.

We structure the problem, then, as a cross-language retrieval problem, with a 'parallel corpus' consisting on one side of bank transactions, on the other of check register transactions, and with the parallelism determined by prior matches.

The algorithm used for reconciliation retrieval is outlined fully in [7]. It is based on Singular Value Decomposition (SVD), a technique used in signal processing for extracting the 'principal components' of data and separating signal from noise. Because, in this case, there are two sources of signal – data from bank statements versus data from a check register, SVD is particularly useful because it compresses the signal into a single stream, allowing the two sources to be correlated with one another. It should be noted that this is exactly the same technique used effectively for cross-language retrieval in [7].

To recap briefly, this algorithm proceeds by 'aligning' terms (words) from the parallel corpus. This is illustrated for machine translation in [6] (p. 267) as follows:

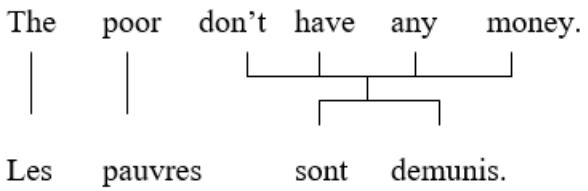

Figure 1: Alignment of terms in a parallel corpus

Alignment is fully automated and involves finding from among the Cartesian product between terms in one language and terms in the other, and for each term in one language, the term in the parallel document which maximizes the pairing's mutual information, which for two terms A and B is calculated as follows [6]:

$$I(A;B) = \sum_{b \in B} \sum_{a \in A} p(a,b) \, log \left(\frac{p(a,b)}{p(a) \times p(b)} \right) \qquad (1)$$

In the case of reconciliation, the ‘parallel corpus’ is previously reconciled transactions, and the ‘words’ in this parallel corpus are transaction descriptors; instead of matching, for example, ‘poor’ and ‘pauvres’, we might be matching the bank’s descriptor ‘ATM’ to the check register descriptor ‘cash’.

It should be noted that in the present case, rather than using the dates themselves as descriptors – which would be of little use, since dates from one month per se cannot be used to train any machine learning algorithm for the next month – we use features involving a probabilistic lag between the dates of a pair of reconciled transactions (see Figure 2 for the distribution based on the actual data we will use in section 3).

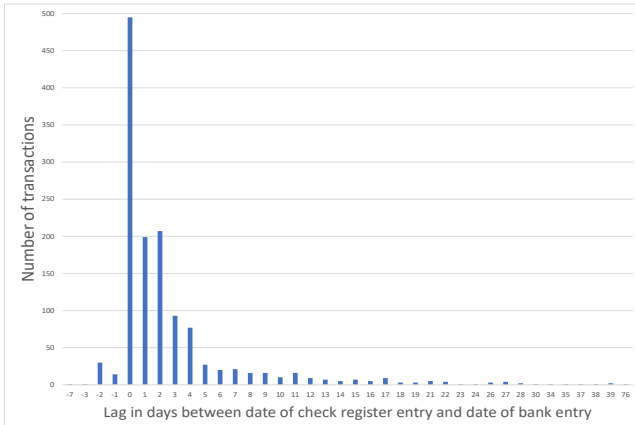

Figure 2: Chart of number of transactions by lag between check register and bank date

From the above, we can construct a sparse matrix encoding alignments, illustrated (for machine translation) in Figure 3 (here, the character ‘?’ in each cell denotes the measure of mutual information between the two respective terms).

	the	poor	don’t	have	any	money	les	pauvres	sont	démunis
the	?						?			
poor		?						?		
don’t			?						?	?
have				?					?	?
any					?				?	?
money						?			?	?
les	?						?			
pauvres		?						?		
sont			?	?	?	?			?	
démunis			?	?	?	?				?

Figure 3: Cross-language term alignment

So far, the techniques described come from machine translation, but the next step – which is from cross-language information retrieval – is to construct a separate ‘term-by-document matrix’ from the aligned data. In

cross-language information retrieval, the terms are words in either language (say, French and English), and the documents are the concatenation of parallel text, e.g. 'I speak with you. Je parle avec toi.'). The cells in the matrix contain weighted frequencies of the given term in the given document.

A matrix multiplication is performed between the alignment matrix and the term-by-document matrix. This allows patterns of co-occurrence between terms to be reinforced by the mutual information from the alignments. The product is then factorized using SVD [9]. SVD factorizes a matrix X as follows:

$$\boldsymbol{X} = \boldsymbol{USV}^T \qquad (2)$$

where U is n orthonormal matrix of left singular term vectors, S is a diagonal matrix of singular values, and V is an orthonormal matrix of right singular document vectors [10]. SVD, also known as Principal Component Analysis, does two things that are important for our purposes: first, it effectively both distills the factorized matrix into its 'principal components' of signal. Secondly, the 'U' matrix resulting from the factorization is a collection of vectors encoding the semantic properties of each term in machine readable format.

The process of performing cross-language information retrieval then involves taking (1) a new document, one not in the parallel corpus (known in information retrieval as the 'query'), and (2) all the existing documents that form the population from which retrieval candidates are to come. Either or both of (1) and (2) can come from either language. Each document in this pool is then also 'projected' int the same 'semantic space' as the parallel corpus by converting it into a vector. This is accomplished by creating a vector for the new documents just as we did for the parallel corpus (determining the terms in the document and their weighted frequencies), and then the 'projection' is achieved by multiplying that vector by the U matrix above. One of the properties of SVD is that these vectors will be in an orthonormal basis, i.e., the principal components or dimensions of the vector are orthogonal to one another – in other words, in a Euclidean multi-dimensional space. This means that the 'similarity' of any vector to any another, say the vector for the query or the vectors for the documents to be retrieved, can easily be computed be calculating the cosine between the vectors. Therefore, to find the most 'relevant' document to a query – its nearest neighbor – one need only compute all the pairwise cosines and find the pair for which the cosine is highest.

Applying all this to the problem of reconciliation boils down to the following: we enumerate all possible alignments of attributes of transactions, such as words used in transaction descriptions; we compute the mutual information of pairs of attributes; we align the attributes by ruling out all but those pairs that maximize mutual information. We then compute a transaction-by-attribute matrix (instead of a term-by-document matrix), weighting it and multiplying it by the alignment matrix just as we did above for the term-alignment and term-by-document matrices. We factorize that product to obtain an attribute-by-principal-component matrix; and then we project new transactions from both the bank and the check register into the 'semantic space' by multiplying by U. Then, to reconcile pairs of transactions, we can use each new transaction from the bank statement as a 'query' and we 'retrieve' the most similar transaction – the nearest neighbor – from the pool of unreconciled check register transaction vectors.

2.2 Integrating AI and the human in a workflow

Now, the question of what constitutes the 'parallel corpus' in our case requires that we turn to discussing how the proposed approach can be set up within an overall workflow; this is itself of interest from a research and

operational point of view, as it touches on 'human factors' (the place of the human within the loop of an AI system).

As already mentioned, we can postulate matches even without any human intervention or prior 'parallel' data based on when monetary amounts uniquely match. For each new month, the process looks as shown in Figure 4. It is important to note here that the flowchart – though often associated with rules-based systems – should not be construed to have those associations here: crucially, all algorithms in automated steps (those shown in blue) are fully unsupervised.

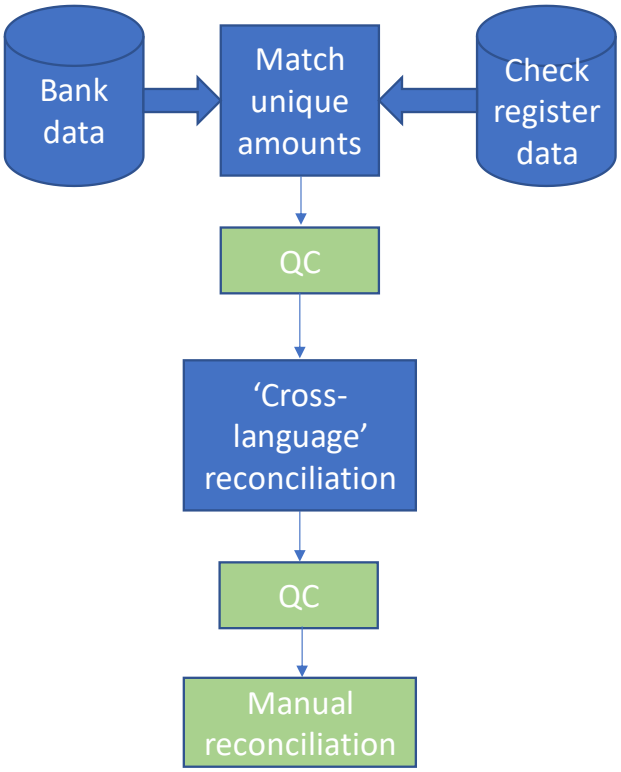

Figure 4: Monthly process

Here, block arrows indicate data ingestion, while regular arrows indicate chronological process flow. Blue rectangles indicate automated processes, while green rectangles indicate processes where a human is involved.

Walking through this process: each month, when the monthly bank statement becomes available, it is downloaded and ingested along with check register data into the reconciliation system. The system first automatically reconciles any transactions where the uniqueness of the monetary amounts precludes more than one possibility for reconciliation. The system-created matches are presented to a human for quality control (QC). (This is desirable, because even where amounts are unique, it is always possible that uniqueness was simply the result of timing: an amount of $153.96 might happen to be unique now, but no longer unique when more data becomes available upon the next ingestion.) QC can be configured for the human either to review

every match made by the machine, or, for large datasets, a sample of matches; the sample size itself can be increased or decreased depending on overall tolerance of risk of 'false positives' (system-reconciled transactions that were inappropriately matched) versus labor-intensiveness. (The larger the sample, the less risk of false positives but the more labor-intensive the process.) During this QC process, the human is given the opportunity to accept or reject each match. Those that are rejected are returned to the pool of unreconciled transactions.

In the next step of the monthly process, the machine makes a second attempt to reconcile any unreconciled transactions, including those that were just returned to the pool through rejection in QC, but also those for which a reconciliation could not be found through unique-amount matching. This is the step that is based on machine learning, and specifically cross-language retrieval, where the parallel corpus comprises everything reconciled (and not rejected) to date, including all of the following:

1. Unique-amount matches from the current month and all prior months;
2. 'Cross-language' matches from all prior months;
3. Manual reconciliation from all prior months (see below).

The key point here is that only those matches that have either been performed by the human in the first place, or reviewed and accepted by a human, are allowed to inform the machine learning algorithm. This underlines the importance of QC: it is a means not just to weed out current errors, but also to improve the quality of all prospective (future) reconciliation. Just as in a supervisor-supervisee relationship, QC serves a training as well as a quality function.

Continuing on, the next step is a second round of human QC, so that 'cross-language' reconciliation can also be controlled for quality. This, again, can be on either a sample basis or for the whole population, depending on tradeoffs between tolerance of risk and labor-intensiveness.

Finally, there may always be some transactions that could not be reconciled by the system – perhaps they were too anomalous in relation to available training data, or because transaction descriptors were previously unencountered or otherwise insufficient to allow a match. The final step is to allow the human to attempt to reconcile these transactions manually. It should be noted here that the difficulty for a human in reconciling 5 remaining pairs of transactions is likely to be well under 5% of the difficulty of reconciling 100 pairs, for the same reason that completing the last 5 pieces of a 100-piece jigsaw puzzle is much easier than piecing together the first 5 pieces: the combinatorics are much less prohibitive. Essentially, the division of labor between the system and the human leaves to the human the part that is least suited for machine learning, at a point where the cognitive load on the human is brought to an absolute minimum.

This whole process, of course, repeats in each new month where the last month's process left off.

To summarize: each automated process is subject to human review, so that if any spurious matches are made by the computer, the human can reject these. This is important, because (again) as with a jigsaw puzzle, spurious 'fits' can prevent the correct pieces from finding where they truly belong, and in the case of the current machine learning algorithm, a spurious match will also cycle back into the training data, reinforcing bad patterns and leading to further similar spurious matches. Each time the human validates matches made by the AI algorithm, on the other hand, these are cycled back into the training data to improve future performance. For each month, there are two quality-control steps, one for the initial matches based on unique amounts, and a second for those matches where the amounts were not unique but the 'cross-language retrieval' approach was

used. The final step for each month's data is for the human manually to 'mop up' any transactions that the computer was unable to match, perhaps because it lacked sufficient aligned training data with similar examples. As with SMT, matches made manually in this way also get cycled back into training data, so that the learning algorithm can, one hopes, identify future similar examples and match them automatically. If the AI system improves and human confidence in the system reaches a sufficient level, it may be possible to scale back QC to be done either on a sample basis, less frequently, or not at all. The calculus for this will also depend on the perceived costs of errors, and this in turn will depend on the specific application.

Note that the above is a monthly process, corresponding to what happens in practice in many operational settings in finance departments.

To recap on what we have outlined in this section: we structure the reconciliation problem as a machine translation / cross-language information retrieval problem. (The mechanics are the same as outlined in other publications, to which we refer the reader.) Since the actual data elements (transaction dates, monetary amounts, values in data fields denoting transaction types, words in transaction descriptions, etc.) are simply treated like 'words' in a 'language', it will be evident that nothing in this approach is tailored to particular data formats, types of reconciliation, or individual datasets: it is truly unsupervised from a machine learning point of view, just as SMT is unsupervised. In this sense, the approach is robust: it need not be reconfigured for new problems, or if data streams suddenly change, perhaps because the bank changed the data format or a new clerk began entering check register descriptions in her own idiosyncratic way. All this is handled through the monthly cycle of QC and machine learning, and the AI system simply adapts to new data as it comes in.

3 EMPIRICAL VALIDATION

We set the above process up using the author's own check register and monthly bank statements, downloaded from the bank's website in csv format, for a period from December 1, 2014 through February 29, 2020 – 51 months with a total of 1,314 transactions, for an average of just under 26 transactions per month. This is undoubtedly a small scale compared to the volume that would be encountered in many industrial settings, but the purpose here is to demonstrate the feasibility of our approach, and we will discuss as we proceed what adjustments might need to be made to scale the approach up.

One configurable setting has to do with the sensitivity of the analysis – in Singular Value Decomposition, the number of principal components returned. A larger number makes the analytics more sensitive to subtle patterns of signal, a smaller number focuses on just the high-level patterns. It is not necessarily the case that greater sensitivity leads to greater accuracy, because a more sensitive analysis also picks up more noise.

We measure how 'good' the automated reconciliation is based on two validation metrics, precision and recall, also widely used in information retrieval [11], and adapted here as follows:

$$precision = \frac{|\{accepted\ \} \cap \{reconciled\ \}|}{|\{reconciled\ \}|} \quad (3)$$

$$recall = \frac{|\{accepted\ \} \cap \{reconciled\ \}|}{|\{accepted\ \}|} \quad (4)$$

F_1 is a composite measure of precision and recall – the harmonic mean between the two [11]:

$$F_1 = 2 \cdot \frac{precision \cdot recall}{precision + recall} \quad (5)$$

Stating this in English, precision means the proportion of automatically reconciled transactions that were accepted by a human in QC. Recall means the proportion of transactions that should have been reconciled, that were actually automatically reconciled. (Assuming that all transactions should end up reconciled, we may assume that the denominator of recall is the total number of transactions to date.)

The higher precision is, the fewer mistakes the machine is making, and the higher recall is, the fewer transactions the machine is unable to reconcile. One can come at the expense of the other: for example, a machine that eschews all but the easiest matches might trivially achieve 100% precision, but be of little use because it still leaves most of the work to the human. This is, essentially, like the problem that exists in much of management: weighing the risk of delegating tasks to a less-experienced subordinate (here, the computer) against the benefit of freeing up the manager's (here, the human's) time. In the current case, it may be possible to adjust certain 'dials' to take into account risk tolerance, and one of those dials might, as already mentioned, be the number of principal components.

First, we should note that in the experiment performed, precision of the first automated reconciliation process (by unique amounts) was 100% across the board. From a machine learning point of view, this process is not interesting in itself, and the result of 100% accuracy is somewhat trivial. This is because almost by definition, the uniqueness of the amounts within a large population (note: we test for uniqueness across all months of data, not just in the current month) is what allows near-certainty to be gained that the match is not spurious. It is therefore unsurprising that the human reviewer always agreed. We remind the reader, however, that the unique-amount matching, while uninteresting per se, is key to 'seeding' the AI that allows reconciliation of the more difficult cases where there may be more than one way potentially to match up transactions. In other words, the trivial process and results of matching unique-amount transactions lead to something of interest: the ability of a machine learning to make subsequent non-trivial inferences from data.

In Table 1, which shows the precision and recall for the machine-learning part (the 'cross-language retrieval'), we see that in general, more principal components (up to the maximum of 40 tested) is unambiguously better: greater sensitivity particularly benefits recall, with no loss (in fact, a small gain) in precision.

Table 1: High-level comparison of performance with various settings

SVD principal components	Precision	Recall	F_1
20	88.0%	81.2%	84.4%
30	88.5%	83.0%	85.6%
40	88.8%	84.1%	86.4%

Figure 5 gives a different view of the results, using the setting of 40 principal components. Here, we see precision over time. At the left side of the chart, it is notable that precision starts at 80%, but over the course of two months, reaches 100% and remains at that level for three months. We believe this can be attributed to the strength of unsupervised learning, its ability to learn from more data. Essentially, just as an inexperienced clerk might need to learn from the correction of a supervisor, the proposed approach learns from the corrections made by the human reviewer 'in the loop' through the QC process.

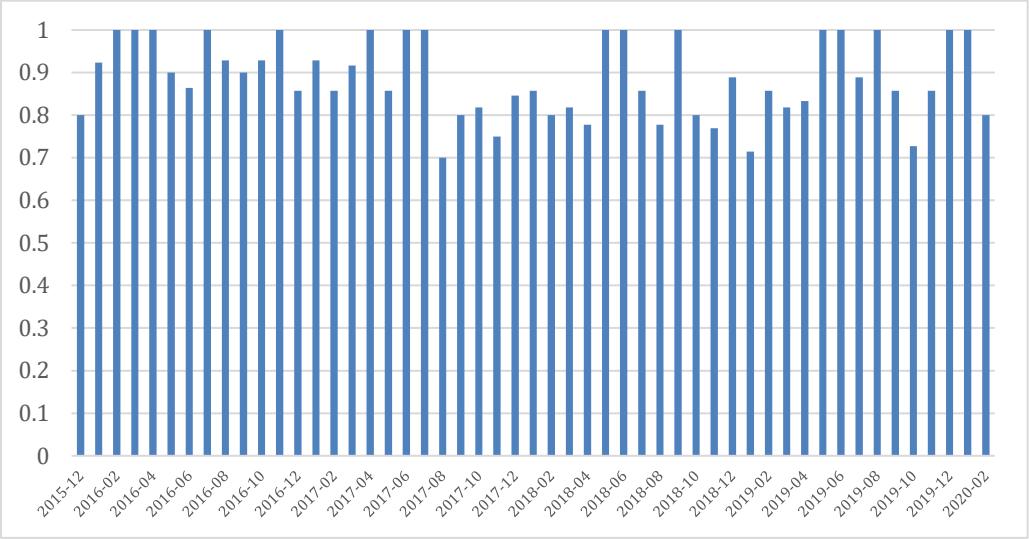

Figure 5: Precision over time, 40 principal components

Why, then, are there some months in which precision drops off? One of the most noticeable examples is for the month 2017-08, where precision suddenly drops to 70% from its prior level of 100%. The answer is that in this month, there were some combinations of transactions that departed from patterns observed before that, and in addition, there were similar transactions with identical amounts close together in time. Such changes in transactional patterns are to be expected from time to time in the ordinary course of business for any entity. As already mentioned, the idea behind SVD (and unsupervised learning in general) is that it picks up on patterns in the data, and so it is possible to see in this case how prior patterns could have turned out to be misleading. As is also evident from the chart, it took a number of months for the AI to 'recover' here. While there are potentially ways in which our algorithm could be improved in such instances (e.g., biasing it towards more recent transactions, being more proactive in 'disciplining' it when errors are made), we will leave these for future research.

4 CONCLUSION

In this paper, we have shown that financial reconciliation, a time-honored staple of the audit function, has adapted to the age of big data, but in a way which is likely to bring new problems. Financial technology professionals have not yet learned the lessons that have been hard-won in another field, machine translation, in particular the lesson that big data necessitates new data-centric ways of thinking about analytics. Despite early optimism, rules-based approaches to machine translation caused efforts quickly to founder, a state which persisted for many decades because the approaches did not properly scale up. In the 1990s, however, researchers realized that big data could actually be turned to the advantage of machine translation, by leveraging the strengths of unsupervised techniques, and the early optimism that machine translation would soon be a 'solved problem' was finally realized, albeit much later than had originally been anticipated.

The good news is that financial reconciliation can be cast in almost identical terms to machine translation, as a 'cross-language information retrieval' problem. The contribution of this paper is to propose this new way of

thinking about the problem – something we do not believe has been done before in the literature – and to demonstrate empirically its promise on a small scale. With a dataset of modest size, we were able to demonstrate initial precision and recall in the mid-to-high-80% range, highly respectable results considering that our approach required no configuration specific to the particular dataset tested. We outlined certain purely computational settings (such as sensitivity or number of principal components) that could be adjusted, and might indeed need to be adjusted for larger datasets or if the risk associated with errors is greater.

But it should be noted, too, that the proposed approach is not purely a computational approach, but rather one which considers the relative places of the human and the machine in the loop, one where, in the end, the entire-system accuracy with quality-control is 100%. The point of this is in our overall 'system', we envisage the human and the machine working together, but dividing up roles and responsibilities to greatest effect, maximizing the strengths and appropriately using the time of each. The result causes the machine-human relationship, in effect, to mirror the supervisee-supervisor relationship, with a good supervisee learning through the supervisor's correction of mistakes. At the same time, we have showed how the process of the supervisor actually correcting these mistakes can seamlessly be incorporated into the larger machine-learning system. We believe consideration of the 'human in the loop' is an indispensable, yet often-forgotten, best practice for the development and implementation of Artificial Intelligence systems in the realm of finance, as in any other domain, and we hope that our work will point the way towards innovation in areas other than financial reconciliation: the time is ripe for it.

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